

Fourier-Based Decoupling Network for Joint Low-Light Image Enhancement and Deblurring

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Abstract—Nighttime handheld photography is often simultaneously affected by low light and blur degradations due to object motion and camera shake. Previous methods typically design specific modules to restore the degradations in the spatial domain independently. However, the interdependence of low light and blur degradations in the spatial domain makes it difficult for these approaches to effectively decouple the degradations, limiting the performance of the designed modules. In this paper, we observe that in the Fourier domain, low light and blur degradations can be represented independently in the amplitude and phase of the image. Through an in-depth analysis of the underlying physical degradation process, we discover that low light degradation exhibits distinct characteristics across different frequency bands in amplitude, while blur degradation is characterized by phase correlation. Leveraging these insights, we mathematically derive a frequency attention mechanism and a filtering mechanism for learning decoupled representations of these degradations, proposing a Fourier-based Decoupling Network for joint low-light image enhancement and deblurring. Experimental results demonstrate that our method achieves the state-of-the-art performance on both synthetic and real-world datasets and exhibits significantly sharper edges. Code is available at <https://github.com/Jabruson/FDN-TIP2025>

Index Terms—Multiple degradations, low-light image enhancement, image deblurring, physical modeling model.

I. INTRODUCTION

IN NIGHTTIME photography, images are often captured in low-light environments. To compensate for insufficient lighting, long exposure techniques are typically employed.

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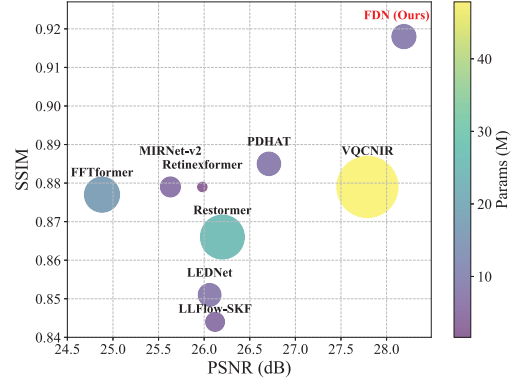


Fig. 1. Improvement of FDN over some SOTA approaches. The size of circle is proportional to the number of the parameters.

However, despite the increased exposure time to improve luminance, images still suffer from problems such as detail loss and color distortion. Moreover, due to camera shake and object motion during photography, long exposure exacerbates blur degradation. Consequently, nighttime photography is often plagued by the dual degradations of low light and blur.

Currently, considerable attention has been paid to low-light image enhancement (LLIE) and image deblurring individually. However, few works address both degradations simultaneously. A straightforward approach is to sequentially apply LLIE and deblurring methods. However, such an approach does not yield satisfactory results. The LLIE process primarily focuses on improving luminance and reducing noise. Hence, if LLIE precedes deblurring, it may result in the loss of high-frequency information, such as textures and edges [1], [2], [3], which are crucial for effective deblurring [4], [5]. Conversely, if deblurring is performed before LLIE, saturated pixels in high-luminance areas may render general deblurring methods ineffective [6], [7], and noise further complicates the accurate estimation of the blur kernel [8]. To this end, ensuring each stage maintains its effectiveness in the cascade process without compromising the restoration performance of the subsequent stage is challenging. A more sophisticated approach is required to address joint LLIE and deblurring.

As a pioneering work, Zhou et al. [1] introduced the LOL-Blur dataset, which contains paired images depicting scenes under low-light blurred and normal-light sharp conditions, providing a necessary dataset for the end-to-end network training. VQCNR [3] is a non-decoupling-based approach, which utilizes a vector quantization codebook to provide high-quality priors for the restoration. However, it is crucial to

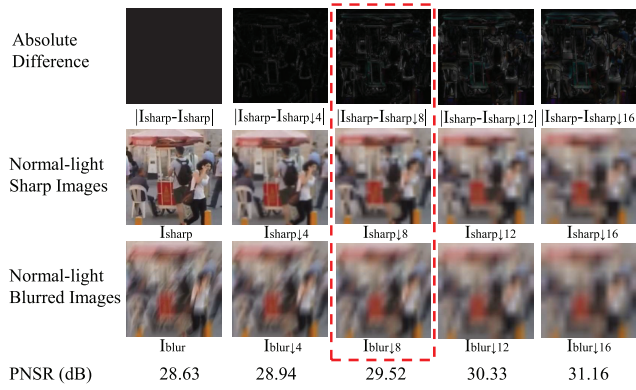


Fig. 2. The results of a pair of normal-light sharp and normal-light blurred images downsampled by multiple factors. The downsampled images are obtained by applying an x -fold downsampling and subsequently performing an x -fold upsampling, which serves to align the original resolution, denoted $I_{sharp,x}$ and $I_{blur,x}$. We also present the difference between normal-light sharp image at the original scale and its various downsampling versions in the first row. The downsampling factor used in [1] and [2] is indicated by the red dashed line.

utilize specific features to restore degradations pertinently [9]. Therefore, VQCNIR imposes constraints on effectively restoring images with sharp edges. In contrast, supervised decoupling-based approaches are proposed in LEDNet [1] and PDHAT [2]. LEDNet is characterized by an encoder-decoder architecture, where the encoder focuses on LLIE and the decoder focuses on image deblurring. PDHAT employs a three-branch structure, with the main branch supported by two auxiliary sub-branches: one for reconstructing high-frequency information and the other for roughly restoring color. It is worth noting that due to the absence of normal-light blurred images for intermediate supervision in the LLIE process, these methods face challenges in decoupling the degradations. Both LEDNet and PDHAT utilize the downsampled normal-light sharp image to supervise LLIE, assuming it approximates the downsampled normal-light blurred image. Downsampling can suppress some blur-related high-frequency content. As shown in the Fig. 2, we computed the PSNR for a pair of normal-light sharp and blurred images under various downsampling factors. We also visualize the information loss induced by downsampling. While a higher downsampling factor increases their similarity, it also amplifies information loss. At a high downsampling factor (16-fold), the sharp image suffers excessive loss in luminance information and detail, rendering it an unreliable supervision for the LLIE stage. Even at the recommended 8-fold factor [1], [2], the downsampled blur and sharp images remain noticeably different. Although PDHAT introduces high-frequency information modeling compared to LEDNet, relying solely on high-frequency information hardly restores sharp edges in complex regions [10], [11]. Moreover, manually setting the range of high frequencies is also challenging.

In contrast, the Fourier Transform offers a more effective approach to decoupling degradations. In the Fourier domain, phase primarily encodes structural information [12], [13], [14], while amplitude captures luminance, contrast, and other statistical properties [15], [16]. However, most prior studies [17], [18], [19], [20], [21], [22] focus on whether degradations manifest in amplitude or phase, without investigating

how degradation features map to specific frequency bands or explaining the underlying physical mechanisms.

In this work, we analyze the manifestations of low light and blur degradations in the Fourier domain and model the physical degradation process via amplitude and phase components. Building upon that low light and blur degradations can be decoupled in the Fourier domain (Fig. 3), we propose a Fourier-based Decoupling Network (FDN) to implement LLIE and deblurring simultaneously. FDN comprises a Multiscale Amplitude Restoration module (MAR) and a Fourier-based Decoupling Transformer (FDformer), where MAR produces coarse amplitude priors for FDformer. Based on our analysis that low light degradation exhibits distinct characteristics across various frequencies in amplitude and blur degradation is characterized by phase correlation, we introduce a Fourier-domain self-attention mechanism to learn separate representations for different degradations. Additionally, recognizing that different frequency bands contain unique information in both amplitude and phase, we further integrate a filtering mechanism to amplify or attenuate specific frequency components of the features. Finally, FDN supports luminance customization by adjusting the coarse amplitude for the need of personalized restoration. The main contributions of our paper are summarized as follows:

- Based on the analyses that the joint degradations of low light and blur can be decoupled in the Fourier domain, we design FDN according to the distinct physical properties of the amplitude and phase components. Our method explores the relationship between phase correlation and blur information, as well as the representation of amplitude modulation and low light information, providing an effective method for joint LLIE and deblurring.
- We design a self-attention mechanism to extract different types of degradation representation and an efficient Feed-Forward Network (FFN) to adaptively learn frequency features in both amplitude and phase. Additionally, we introduce a Fourier-based cross-attention mechanism to provide crucial priors for frequency learning.
- Experimental results show that our method achieves SOTA performance with only 16.7% parameters compared to the current SOTA method. Especially, our restoration results exhibit substantially superior performance in restoring edges.

The structure of this paper is outlined as follows. Sec. II delves into the related work concerning image restoration. Sec. III provides a detailed introduction to the FDN. Sec. IV presents comprehensive experimental results alongside corresponding analyses. Finally, Sec. V wraps up the paper, offering conclusions and potential directions for our future work.

II. RELATED WORKS

A. Low-Light Image Enhancement

LLIE methods are dedicated to improving the quality of images captured under low-light conditions, primarily focusing on improving luminance, increasing contrast, reducing noise, and restoring details.

With the advancement of deep learning in computer vision [23], [24], many learning-based approaches have been integrated into LLIE [15], [25], [26]. Lore et al. [27] pioneered the

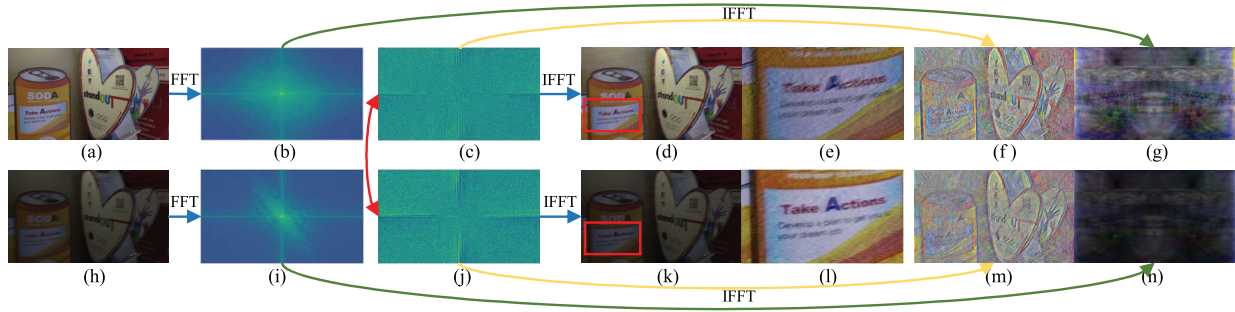


Fig. 3. Analysis of the decoupling process based on phase and amplitude in the Fourier domain. By swapping the phase components between a normal-light sharp image (a) and a low-light blurred image (h), we obtain a normal-light blurred image (d) and a low-light sharp image (k) (zoom in for a better view). Additionally, we visualize the Inverse Fourier Transform of phase and amplitude components to illustrate their distinct representation.

application of deep learning methods for LLIE. Wu et al. [28] introduced semantic priors into their LLIE model, resulting in enhanced results with natural textures and plausible colors. Cai et al. [29] proposed a single-stage LLIE method based on the Retinex theory. The existing spatial domain-based methods often neglect the Fourier frequency information, which encapsulates global attributes and is capable of disentangling image degradation and content component [21].

In recent years, there has been a growing interest in exploring the utilization of Fourier frequency information in LLIE [15], [16]. These methods primarily focus on the influence of amplitude on luminance. However, they often overlook the impact in scenarios with blur.

B. Image Deblurring

Deblurring aims to reduce or eliminate image blurring caused by factors such as motion blur, lens blur, or system misalignment during image acquisition.

Nah et al. [30] proposed an end-to-end CNN for deblurring, directly obtaining sharp images in a multiscale architecture without estimating the blur kernel. Inspired by this work, many works based on multiscale [31], [32] and multistage [33], [34] approaches have been applied to deblurring, with promising results. However, deblurring often requires consideration of global information, which is a limitation of CNNs. While transformers excel in establishing long-range dependencies, Yan et al. [35] introduced a transformer architecture to encapsulate a spectrum of global and local information, thereby facilitating the restoration of significant blur variations.

Considering that frequency information can provide an effective prior for deblurring, Mao et al. [36] achieved the separation of blur information by applying ReLU operations for frequency selection. Kong et al. [37] developed an efficient frequency domain-based self-attention solver building upon the theory that convolution in the spatial domain corresponds to multiplication in the frequency domain. Nevertheless, in low-light environments, issues such as saturated pixels in high-luminance areas and noise [6], [7], [8] present substantial obstacles for effective deblurring.

C. Joint Solutions

Most current works on image restoration focus solely on single degradation. However, in practical scenarios, images

often suffer from multiple degradations. Simply employing a cascade approach may lead to error accumulation.

In recent years, an increasing number of studies have delved into the realm of joint restoration for multiple degradations. Wan et al. [38] addressed the common occurrence of rain in low light conditions by proposing a spatially adaptive network guided by degradation masks. Lu and Jung [39] introduced two branches to respectively learn restoration coefficients and iteratively perform joint restoration for LLIE and denoising. Zhang et al. [40] proposed an attention dual supervised network, effectively addressing both blurring and low resolution. However, these methods do not consider low light and blur degradations.

Zhou et al. [1] devised specific modules to tackle different degradations in low-light conditions and deblurring scenarios. Building upon this, Li et al. [2] proposed a parallel architecture that decomposes the restoration of low light and blur degradations into high-frequency detail restoration and perceptual restoration. The underlying assumption of these methods is that downsampled blurred images can approximate downsampled sharp images. However, the interdependence of low light and blur degradations in the spatial domain makes it challenging to decouple the degradations effectively. Zou et al. [3] utilized vector quantization codebooks to provide high-quality image priors. However, its non-decoupled nature limits its ability to restore sharp edges.

D. Fourier Frequency Information in Image Restoration

The Fourier Transform enables global spectral analysis and effective separation of frequency components, which has been used in traditional image restoration [41], [42], [43], [44], [45]. Recently, deep learning-based methods further exploit Fourier priors by decoupling images into amplitude and phase components.

Considering that the amplitude component encodes luminance information, Huang et al. [17] decoupled low light degradation in the amplitude domain, using separate amplitude and phase sub-networks for progressive luminance enhancement. Furthermore, Wang et al. [15] exploited the positive correlation between amplitude and luminance to achieve efficient LLIE. Amplitude has also been shown to characterize haze [18], rain [19], and JPEG artifacts [20]. These approaches split deep features into amplitude and phase, with dedicated modules for each. Dual-domain methods further combine

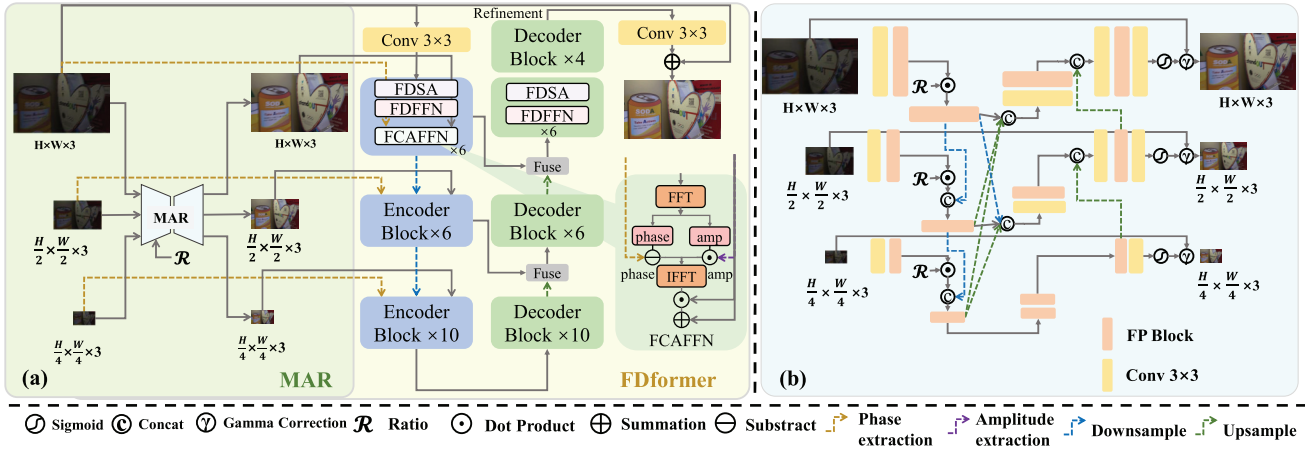


Fig. 4. The architecture of our FDN. (a) is the overall architecture. (b) is the detail of MAR. FDN comprises MAR and FDformer. MAR receives raw inputs at three different scales and outputs corresponding coarse-restored results for each scale. At each of the three scales, the phase of the input image and the coarse-restored amplitude obtained by MAR guide the FDformer to restore the desired results through FCAFFN in the encoder of the FDformer.

spatial and frequency representations for richer features [18], [20]. Given the ability of the Fourier Transform to decouple degradation and content information, Zhou et al. [21] proposed a general model that learns degradation priors from the Fourier domain, demonstrating strong performance across tasks.

While most of the methods primarily exploit the amplitude component, Li et al. [16] demonstrated that luminance information is predominantly represented by the amplitude, whereas noise concentrates in the phase. By independently processing amplitude and phase, the approach achieves effective luminance enhancement and noise suppression. He et al. [46] decoupled panchromatic and multi-spectral features into amplitude and phase, injecting amplitude for high-frequency detail and fusing phase for pan-sharpening. Lv et al. [22] proposed a zero-shot framework for joint LLIE and deblurring by leveraging the generative priors of diffusion models. They decoupled low light and blur degradations into amplitude and phase, enabling the model to learn luminance priors from the diffusion process and iteratively refine phase through a frequency optimization strategy, which is able to effectively handle the degradations without any paired training data.

Despite these advances, existing methods do not model the physical degradation process in the Fourier domain, which limits both interpretability and overall performance.

III. PROPOSED METHOD

A. Overview

Our goal is to develop a decoupling-based modeling network based on the phase and amplitude components in the Fourier domain. Fig. 4(a) illustrates the overall architecture of FDN, which comprises MAR and FDformer. In Sec. III-B, we analyze the manifestations of blur and low light degradations in the phase and amplitude components, respectively. Then we introduce MAR in Sec. III-C, followed by the explanation of three components of FDformer: Fourier-based Decoupling Self-Attention (FDSA) in Sec. III-D.1; Fourier-based Decoupling Feed-Forward Network (FDFFN) in Sec. III-D.2; and Fourier-based Cross-Attention Feed-Forward Network (FCAFFN) in Sec. III-D.3.

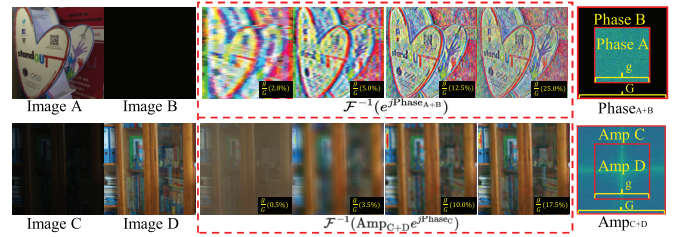


Fig. 5. The characterization of frequency bands of the phase and amplitude, respectively. (A) Normal-light sharp image. (B) Entirely black. (C) and (D) Pair of low-light and normal-light sharp images. The frequency components are processed with spectrum centering.

B. Decoupling of Multiple Degradations in Fourier Domain

As shown in Fig. 3, we select two images, a low-light blurred image and a normal-light sharp image. After the Fourier Transform, we exchange their phases, resulting in a low-light sharp image, and a normal-light blurred image. It suggests that phase primarily encodes structural information, while amplitude mainly encodes luminance information. Other images exhibit similar characteristics in our experiment.

We further investigate the characterization of frequency bands (Fig. 5) of the phase and amplitude respectively. In the first row, only the low-frequency (central) phase of Image A (Phase A) is retained, and the proportion of Phase A, denoted as $\frac{g}{G}$, is increased. When the proportion of Phase A is low, the structure of A appears as a vague outline. As the proportion increases, the structural details of A become progressively clearer and more complete. The observation indicates that high-frequency phases primarily encode edge information, while low-frequency phases predominantly represent the overall image structure. In the second row, the first combination result is obtained by replacing the DC component $|F(0,0)|$ of Image C with $|F(0,0)|$ of Image D. As a result, the combined image exhibits similar overall luminance to Image D but with low contrast. As the proportion of Amplitude D increases, the color and contrast in detailed areas gradually improve, approaching those of Image D. This indicates that low-frequency amplitude primarily encodes the overall

luminance, while high-frequency amplitude encodes the luminance in detailed regions.

1) *Luminance Information in Amplitude*: For a grayscale image $f(x, y)$ with dimensions $H \times W$, its amplitude component $|F(0, 0)|$ is given by $\sum_{x=0}^{H-1} \sum_{y=0}^{W-1} f(x, y)$, which is equivalent to the summation of all pixels value in $f(x, y)$. We regard $\frac{1}{HW} |F(0, 0)|$ as the global luminance $\mathcal{L} \in \mathbb{R}^1$ of $f(x, y)$, which is also a widely used metric for overall luminance [47], [48], [49]. Similarly, for an RGB image, we can convert it to grayscale and then obtain $\mathcal{L}$. Thus, we can increase $|F(0, 0)|$ to improve the luminance in $f(x, y)$.

To investigate the impact of $|F(0, 0)|$ on the $f(x, y)$, we change $|F(0, 0)|$ to $\beta |F(0, 0)|$, where β is an amplification factor ($\beta > 1$). The new image $f'(x, y)$ equals to $f(x, y) + (\beta - 1) \frac{|F(0, 0)|}{HW}$. By scaling the component $|F(0, 0)|$ to $\beta |F(0, 0)|$, the global luminance of $f(x, y)$ becomes β times its original value. However, since this operation uniformly increases every pixel value without accounting for the color distribution, it induces color distortion.

Accounting for the color distribution, $f(x, y)$ can be scaled by an amplification factor β to $\beta f(x, y)$. According to the linearity property of the Fourier Transform, we can infer that $\mathcal{F}(\beta f(x, y)) = \beta \mathcal{F}(f(x, y))$, where $\mathcal{F}$ is the Fourier Transform and $F(u, v) = \mathcal{F}(f(x, y))$. Consequently, the amplitude component becomes $\beta |F(u, v)|$ while the phase component remains unchanged. By scaling the entire amplitude spectrum $|F(u, v)|$ rather than scaling only $|F(0, 0)|$, the global luminance can be adjusted while preserving color fidelity.

2) *Blur Information in Phase*: Blur degradation primarily manifests as phase distortion in the Fourier domain. Phase-based video motion processing methods [50], [51] have demonstrated the potential of extracting motion patterns from the phase component. Modeling the relationship between the phase component and degradation based on the physical process of blurring can further alleviate feature entanglement in multi-degradation scenarios. Blurring can be modeled by averaging a series of consecutive frames [1]:

$$\hat{S}_{blur} = g \left(\frac{1}{m} \sum_{i=1}^m S[i] \right), \quad (1)$$

where $\hat{S}_{blur} \in \mathbb{R}^{3 \times H \times W}$ represents the observed blurred image, $g(\cdot)$ represents mapping the underlying sharp signal to the observed Camera Response Function (CRF), m denotes the total number of frames within the image-taking period, $S[i]$ denotes the sharp signal in the i -th frame. Following [1], Eq. 1 can be transformed as:

$$\hat{S}_{blur} = \frac{1}{m} \sum_{i=1}^m \hat{S}(\mathbf{x} + \Delta \mathbf{x}_i), \quad (2)$$

where $\Delta \mathbf{x}_i$ represents the displacement of i -th frame relative to the first frame, making $\hat{S}(\mathbf{x} + \Delta \mathbf{x}_i) \in \mathbb{R}^{3 \times H \times W}$ the image in the i -th frame.

Given that the exposure time is typically not excessively long, the difference of adjacent frames $\Delta \mathbf{x}_{i+1} - \Delta \mathbf{x}_i$ results in a small displacement. Following [52], we can consider this process approximates uniform linear motion, with consistent displacement $\Delta \mathbf{x}$ between adjacent frames $\Delta \mathbf{x}_{i+1} - \Delta \mathbf{x}_i = \Delta \mathbf{x}$. Therefore:

$$\hat{S}_{blur} = \frac{1}{m} \sum_{i=1}^m \hat{S}(\mathbf{x} + (i-1)\Delta \mathbf{x}). \quad (3)$$

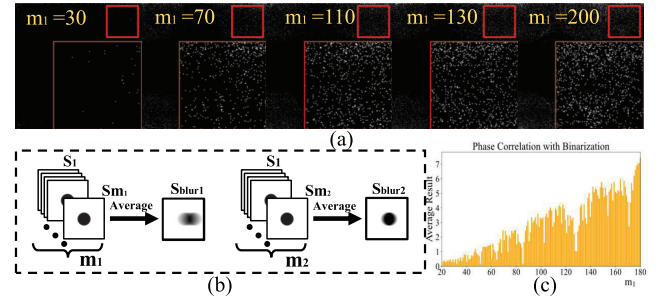


Fig. 6. Visualization of the relationship between phase correlation and blur information. We generated blurred images $\hat{S}_{blur1}$ and $\hat{S}_{blur2}$ by varying the frame count m of identical circles ($m_2 = 20$, m_1 varied). (a) are the results of $\mathcal{F}^{-1}(e^{j(P_{blur1}(\omega) - P_{blur2}(\omega))})$, where all non-zero pixels are binarized to 255 and their count indicates relative blur degree. (b) is the schematic of the blurred image formation. (c) is the bar chart illustrating the variation in the averaged proportion of white pixels as m_1 changes in (a).

Combined with the Fourier shift theorem, we perform a Fourier Transform on Eq. 3:

$$A_{blur}(\omega) e^{jP_{blur}(\omega)} = \frac{1}{m} \sum_{i=1}^m A_i(\omega) e^{j(P_1 + (i-1)\Delta P)}, \quad (4)$$

where $A_i(\omega) \in \mathbb{R}^{3 \times H \times W}$ and $P_i(\omega) \in \mathbb{R}^{3 \times H \times W}$ represent the amplitude and phase, respectively, and j denotes the imaginary unit. $\Delta P = 2\pi \Delta \mathbf{x} \omega$ denotes the shift in phase between adjacent frames and P_1 is the phase of the first frame. We only consider the representation of blurring in the phase, thus, we can set $A_i(\omega)$ as 1 [50], [51]. Using the geometric series summation formula, we can get:

$$e^{jP_{blur}(\omega)} = \frac{1}{m} e^{jP_1} \left(\frac{e^{jm\Delta P} - 1}{e^{j\Delta P} - 1} \right). \quad (5)$$

We consider the phase of two images $blur_1$ and $blur_2$ which share the same initial phase P_1 but differ in the total frames m . m can also be characterized as blur degree. The phase correlation between $blur_1$ and $blur_2$ can be represented as:

$$\begin{aligned} & j(P_{blur1}(\omega) - P_{blur2}(\omega)) \\ &= \frac{1}{2} j\Delta P \Delta m + \ln \left(\frac{m_2}{m_2 + \Delta m} \cdot \frac{\sin((m_2 + \Delta m)\Delta P/2)}{\sin(m_2\Delta P/2)} \right), \end{aligned} \quad (6)$$

where m_2 and $m_2 + \Delta m$ represents the degree of $blur_2$ and $blur_1$, Δm is the difference in blur degree. Since the phases component $P_{blur1}(\omega)$ and $P_{blur2}(\omega)$ are real quantities, the right-hand side of Eq. 6 must be purely imaginary. The Eq. 6 can be simplified to:

$$j(P_{blur1}(\omega) - P_{blur2}(\omega)) = \frac{1}{2} j\Delta P \Delta m + \text{bias}(\omega), \quad (7)$$

where $\text{bias}(\omega)$ is a vector in which each element takes a value of either 0 or $j\pi$, with the specific value depending on the logarithm term in Eq. 6. Considering that ΔP is the phase shift between adjacent frames, Eq. 7 indicates that the phase correlation is proportional to the relative blur degree Δm , potentially offset by a constant $j\pi$. Eq. 7 mathematically describes the relationship between phase component and relative blur degree information. As shown in Fig. 6, the difference in blur degree between $blur_1$ and $blur_2$ is approximately positively correlated

with the number of white pixels in $\mathcal{F}^{-1}(e^{j(P_{blur_1}(\omega) - P_{blur_2}(\omega))})$, which is consistent with our theoretical expectations.

Thus, we can develop our method by decoupling low light and blur degradations in the Fourier domain.

C. MAR

We propose the MAR for coarse amplitudes restoration. The restored coarse amplitudes serve as luminance priors, enabling FDformer to focus on fine detail recovery.

MAR is built upon a U-Net architecture, with FPBlock [15] selected as the feature extraction unit. By separately extracting amplitude and phase information, FPBlock achieves superior global representation compared to conventional convolutions. Considering that gamma correction adjusts each pixel individually, making it more effective in preserving the structure of the image in luminance improvement, MAR learns a pixel-wise correction factor $\gamma_n \in \mathbb{R}^{3 \times \frac{H}{n} \times \frac{W}{n}}$ within the range [0,1] for each scale n . γ_n is applied through gamma correction [53] for the inputs $I^n \in \mathbb{R}^{3 \times \frac{H}{n} \times \frac{W}{n}}$ and obtain $I_{mar}^n \in \mathbb{R}^{3 \times \frac{H}{n} \times \frac{W}{n}}$. The scale factor n is selected based on the resolutions of the features in FDFormer, allowing the amplitude components $|\mathcal{F}(I_{mar}^n)|$ to be injected at the corresponding levels in the encoder of FDformer to serve as priors.

Additionally, since the luminance of the recovered result for a low-light blurred image is not unique, we encode customized luminance into the coarse amplitudes and use them as priors to guide luminance restoration in FDformer. Based on the previous analysis, we can uniformly amplify all pixel values of the image by a certain factor to increase the global luminance while keeping the phase components unchanged. In the intermediate features for each scale n , we introduce a scaling factor $\mathcal{R}$ to modulate the features $x^n \in \mathbb{R}^{C \times \frac{H}{n} \times \frac{W}{n}}$, enabling user-customized luminance in $|\mathcal{F}(I_{mar}^n)|$, where $\mathcal{R} = \mathcal{L}_{ori}/\mathcal{L}_{exp} \in \mathbb{R}^1$ is the ratio of the global luminance $\mathcal{L}_{ori}$ of the original input image and the expected global luminance $\mathcal{L}_{exp}$ as users' expectation, serving as a turnable parameter.

D. FDformer

As shown in Fig. 4(b), FDformer adopts an asymmetric encoder-decoder architecture, which consists of three main components: FDSA, FDFFN, and FCAFFN. In the following sections, we will provide a detailed introduction to the specific composition of each component.

1) *FDSA*: We propose Fourier-based Decoupling Self-Attention (FDSA) that extracts blurred features from the phase component and low light features from the amplitude component to capture distinct degradation representations.

FDSA is inspired by autocorrelation-based attention mechanisms [54], [55], [56]:

$$Att = \mathcal{F}^{-1}(\mathcal{F}(\mathcal{Q}) \odot \mathcal{F}^*(\mathcal{K})), \quad (8)$$

where $\mathcal{Q}, \mathcal{K} \in \mathbb{R}^{\hat{C} \times \frac{\hat{H}}{p} \times \frac{\hat{W}}{p} \times p \times p}$ represent the Query and Key in the self-attention mechanism, respectively. $Att \in \mathbb{R}^{\hat{C} \times \frac{\hat{H}}{p} \times \frac{\hat{W}}{p} \times p \times p}$ is the attention map. $\mathcal{F}(\cdot)$ denotes the Fourier Transform, $\mathcal{F}^{-1}(\cdot)$ is the inverse Fourier Transform, $\mathcal{F}^*(\mathcal{K})$ represents the complex conjugate of $\mathcal{F}(\mathcal{K})$, and $\odot$ denotes the dot product. The patch size is denoted by p . Eq. 8 is also equivalent to:

$$Att = \mathcal{F}^{-1}((\mathcal{Q}_{amp} \odot \mathcal{K}_{amp})e^{j(\mathcal{Q}_{phase} - \mathcal{K}_{phase})}), \quad (9)$$

where $(\cdot)_{amp}$ and $(\cdot)_{phase}$ represent the amplitude and phase components of the variable, respectively.

According to Eq. 7, the phase correlation between two blurred images reflects their relative blur information. Moreover, phase correlation captures the similarity between signals [57], [58], [59], which aligns with the purpose of self-attention in measuring the similarity between queries and keys. Motivated by this, we integrate phase correlation into the self-attention mechanism to better represent blur information. By setting $\mathcal{Q}_{amp} = \mathbf{I}$ and $\mathcal{K}_{amp} = \mathbf{I}$, where $\mathbf{I}$ denotes an all-ones matrix, the general attention formulation in Eq. 9 reduces to a special case in which only the phase component serves as the variable:

$$Att_1 = \mathcal{F}^{-1}(e^{j(\mathcal{Q}_{phase} - \mathcal{K}_{phase})}). \quad (10)$$

It is noteworthy that this transition from the general formulation to the specific case preserves the inherent mathematical properties of the expression, indicating that the phase correlation between $\mathcal{Q}$ and $\mathcal{K}$ exhibits the mathematical characteristics of self-attention. Therefore, Eq. 10 can be regarded as an amplitude-free self-attention mechanism, which is inherently robust to illumination variations [60]. This robustness enables the effective extraction of blur degradation information even under low-light conditions.

To extract low-light degradation information, we introduce a self-attention mechanism based on the amplitude components of $\mathcal{Q}$ and $\mathcal{K}$. Specifically, we regard $\mathcal{K}_{amp}$ as a weighting matrix for $\mathcal{Q}_{amp}$. This operation can be interpreted as treating $\mathcal{K}_{amp}$ as a filter that selectively amplifies or attenuates specific frequency ranges in $\mathcal{Q}_{amp}$:

$$Att_2 = \mathcal{F}^{-1}(\mathcal{Q}_{amp} \odot \mathcal{K}_{amp}). \quad (11)$$

Similarly, when the phase is set to 0, Eq. 9 can be derived to Eq. 11, indicating that Eq. 11 also possesses the mathematical properties for self-attention.

In the Fourier domain, amplitude represents the magnitude of each frequency component in an image, with larger amplitudes indicating a stronger presence of those frequency components. In contrast, phase specifies the displacement of each frequency component from its origin. Computing self-attention using only phase corresponds to shifting plane waves of equal amplitude. Although Eq. 10 preserves structural information, it discards magnitude details, hindering the identification of key feature locations (Fig. 3(f) and Fig. 3(m)). Similarly, only involving amplitude in self-attention fails to represent effective image information (Fig. 3(g) and Fig. 3(n)). Therefore, Eq. 11 cannot yield meaningful self-attention features.

Therefore, we introduce an extra branch $\mathcal{E} \in \mathbb{R}^{\hat{C} \times \frac{\hat{H}}{p} \times \frac{\hat{W}}{p} \times p \times p}$ in Eq. 10 and Eq. 11 to supplement the missing amplitude and phase information. Our method includes three types of self-attention mechanisms for different representation: phase correlation attention (Att_{f1}), amplitude modulation attention (Att_{f2}), and hybrid attention (Att_{f3}):

$$\begin{cases} Att_{f1} = \mathcal{F}^{-1}(\mathcal{E}_{amp}e^{j(\mathcal{Q}_{phase} - \mathcal{K}_{phase})}), \\ Att_{f2} = \mathcal{F}^{-1}((\mathcal{Q}_{amp} \odot \mathcal{K}_{amp})e^{j\mathcal{E}_{phase}}), \\ Att_{f3} = \mathcal{F}^{-1}((\mathcal{Q}_{amp} \odot \mathcal{K}_{amp})e^{j(\mathcal{Q}_{phase} - \mathcal{K}_{phase})}). \end{cases} \quad (12)$$

Att_{f1} is independent of the amplitude of $\mathcal{Q}$ and $\mathcal{K}$, preventing the introduction of features related to low light degradation

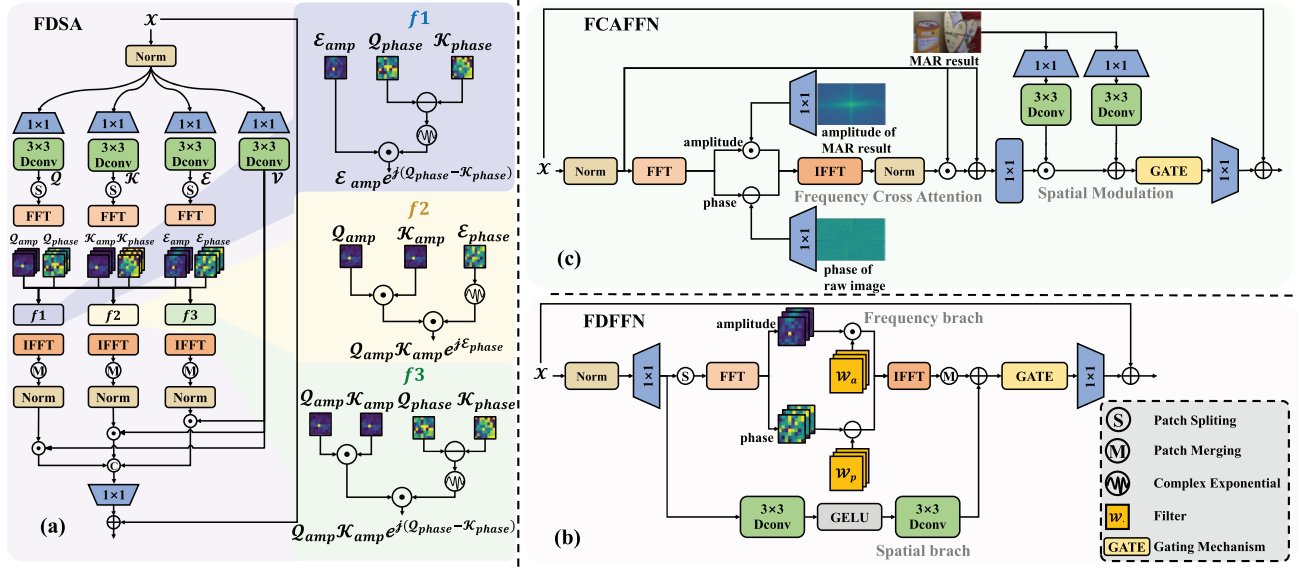


Fig. 7. (a) The proposed FDSA, where $f1$, $f2$, and $f3$ represent three different types of attention: phase correlation attention for representing blur information, amplitude modulation attention for representing low light degradation information, and hybrid attention. (b) FDFFN learns deep frequency domain features. (c) FCAFFN aggregates coarsely restored amplitude and multiscale phase information with FDformer.

while capturing the blur information between Q and K . Similarly, Att_{f2} captures amplitude features related to low light degradation between Q and K while avoiding the introduction of structure-related features associated with Q and K . Finally, Att_{f3} simultaneously considers the attention on both phase and amplitude of Q and K . Our proposed FDSA is illustrated in Fig. 7(a).

2) **FDFFN**: Recognizing that low light and blur degradations can be represented independently in amplitude and phase across different frequency bands (Sec. III-B), we introduce FDFFN to adaptively learn frequency selection and offset in high-dimensional spaces.

As shown in Fig. 7(b), FDFFN consists of parallel frequency and spatial branches. The frequency branch employs learnable filters to perform frequency selection and offset separately on the amplitude and phase of the features. The frequency branch feature $\hat{x}_{fre}^n \in \mathbb{R}^{\hat{C} \times \hat{H} \times \hat{W}}$ is obtained through a patch splitting operation $PS_{p \times p}(\cdot)$ into $p \times p$, and then is adjusted for amplitude and phase information using learnable filters in the Fourier domain. Let $x^n \in \mathbb{R}^{\hat{C} \times \hat{H} \times \hat{W}}$ denotes the proceeded input feature at scale n :

$$\begin{aligned} \hat{x}_{phase}^n &= |\mathcal{F}(PS_{p \times p}(x^n))| - W_{phase}^n, \\ \hat{x}_{amp}^n &= \arctan(\mathcal{F}(PS_{p \times p}(x^n))) \odot W_{amp}^n, \\ \hat{x}_{fre}^n &= PM_{p \times p}\left(\mathcal{F}^{-1}\left(\hat{x}_{amp}^n e^{j\hat{x}_{phase}^n}\right)\right), \end{aligned} \quad (13)$$

where $PM_{p \times p}(\cdot)$ denotes the patch merging. $W_{phase}^n \in \mathbb{R}^{\hat{C} \times \frac{\hat{H}}{p} \times \frac{\hat{W}}{p} \times p \times p}$ and $W_{amp}^n \in \mathbb{R}^{\hat{C} \times \frac{\hat{H}}{p} \times \frac{\hat{W}}{p} \times p \times p}$ are learnable parameters initialized to 2π and 1, respectively, indicating no alteration to phase and amplitude, learning feature modulation from the initial feature space.

In the spatial branch, $\hat{x}_{space}^n \in \mathbb{R}^{\hat{C} \times \hat{H} \times \hat{W}}$ are obtained through two 3×3 depth-wise convolutions and a GELU non-linearity [61] for channel-wise local feature learning.

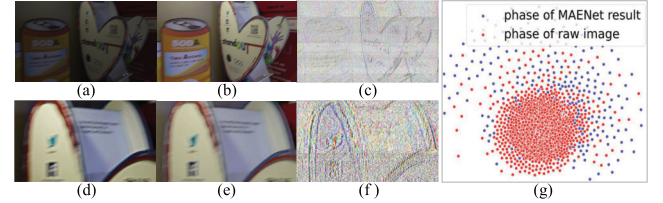


Fig. 8. Display of the result of MAR and explanation of phase information loss. (a) is the original low-light blurred image, (b) is the result after luminance improvement by MAR, and (c) is the difference map of the phase maps of (a) and (b), revealing some differences in edges. (d), (e) and (f) show the zoom in (a), (b) and (c), where (d) is the result of (a) after brightening. (g) illustrates the t-SNE of the result of MAR and the original images in phase, revealing that they are not entirely identical.

After aggregating spatial and frequency features via additive fusion, we introduce nonlinear transformations through gating mechanisms [62].

3) **FCAFFN**: Multiscale blurred features have been proven beneficial for deblurring [11], [31], [32]. To incorporate multiscale phase features and coarse amplitude prior from MAR into the FDformer, we propose the Fourier-based Cross-Attention (FCA) mechanism (Fig. 7(c)), inspired by the cross-attention mechanism and Eq. 12:

$$\begin{aligned} \hat{x}_{phase}^n &= X_{phase}^n - W_{1 \times 1}^n \arctan(\mathcal{F}(I^n)), \\ \hat{x}_{amp}^n &= X_{amp}^n \odot W_{1 \times 1}^n |\mathcal{F}(I_{mar}^n)|, \\ \hat{x}_{cross}^n &= LN\left(\mathcal{F}^{-1}(\hat{x}_{amp}^n e^{j\hat{x}_{phase}^n})\right) \odot x^n + x^n, \end{aligned} \quad (14)$$

where $x^n \in \mathbb{R}^{\hat{C} \times \hat{H} \times \hat{W}}$ represents the feature within FDformer at scale n , $X_{phase}^n \in \mathbb{R}^{\hat{C} \times \hat{H} \times \hat{W}}$ and $X_{amp}^n \in \mathbb{R}^{\hat{C} \times \hat{H} \times \hat{W}}$ are the phase and amplitude of x^n , $LN(\cdot)$ denotes the layer normalize. $X_{phase}^n - W_{1 \times 1}^n \arctan(\mathcal{F}(I^n))$ learns the relative blur degree information between the input image and the features in FDformer. $|\mathcal{F}(I_{mar}^n)|$ is utilized to modulate the amplitude X_{amp}^n . We employ $\arctan(\mathcal{F}(I^n))$ instead of the phase restored

by MAR since we observe that without normal-light blurred images, MAR may lead to the corruption of phase information (Fig. 8).

Moreover, simply combining the amplitude of a normal-light sharp image with the phase of a low-light blurred image may introduce unexpected textures (Fig. 3(e)), indicating the misalignment between the amplitude and phase representations. To assist the adaptive alignment of amplitude and phase, we introduce an additional spatial modulation mechanism:

$$\hat{\mathbf{x}}_{fca}^n = \hat{\mathbf{x}}_{cross}^n \odot \mathbf{W}_{3 \times 3}^n (\mathbf{W}_{1 \times 1}^n (\mathbf{I}_{mar}^n)) + \mathbf{W}_{3 \times 3}^n (\mathbf{W}_{1 \times 1}^n (\mathbf{I}_{mar}^n)), \quad (15)$$

where $\hat{\mathbf{x}}_{fca}^n \in \mathbb{R}^{nC \times \frac{H}{n} \times \frac{W}{n}}$ denotes the output feature of FCAFFN at scale n .

E. Training Strategy

Our FDN adopts a two-stage training process. We initially train MAR, then freeze its parameters, and proceed to train FDformer. The outputs of MAR, denoted as the amplitude prior, are incorporated into FCAFFN in FDformer. During the training process of MAR, we introduce two loss functions: (1) coarse alignment loss L_{coarse} and (2) amplitude loss L_{amp} .

1) *Coarse Alignment Loss*: MAR aims to provide coarse amplitude concerning luminance and contrast. Due to the lack of normal-light blurred images as intermediate GT during training, we introduce a more flexible supervision method instead of pixel-to-pixel supervision:

$$L_{coarse} = \sum_{n=1}^3 \left\| \text{Avg}_{1/2^n}(\mathbf{I}_{mar}^n) - \text{Avg}_{1/2^n}(\mathbf{I}_{gt}^n) \right\|_2, \quad (16)$$

where $\mathbf{I}_{mar}^n$ represents the output of MAR at the n -th scale. $\text{Avg}_{1/2^n}$ denotes average pooling downsampling by n times. $\mathbf{I}_{gt}^n$ is the GT of the same scale as $\mathbf{I}_{mar}^n$.

2) *Amplitude Loss*: Given that MAR primarily focuses on providing coarse amplitude restoration, we introduce an additional loss function that directly supervises the amplitude:

$$L_{amp} = \sum_{n=1}^3 \left\| |\mathcal{F}(\mathbf{I}_{mar}^n)| - |\mathcal{F}(\mathbf{I}_{gt}^n)| \right\|_2, \quad (17)$$

where $|\mathcal{F}(\mathbf{I}_{gt}^n)|$ denotes the amplitude of $\mathbf{I}_{gt}$ at scale n . The overall loss function during the training phase of MAR can be formulated as:

$$L_{stage1} = L_{coarse} + \lambda_1 L_{amp}, \quad (18)$$

where λ_1 is set to 0.01 for balancing the two losses.

The loss function of FDformer consists of the following three parts (1) pixel loss L_{pix} , which is equivalent to L1 loss, (2) frequency loss L_{fft} , and (3) perceptual loss L_{per} .

3) *Frequency Loss*: Frequency Loss is used to supervise the spectrum of the output result in the Fourier frequency domain:

$$L_{fft} = \left\| \text{Real}(\mathcal{F}(\mathbf{I}_{out})) - \text{Real}(\mathcal{F}(\mathbf{I}_{gt})) \right\|_1 + \left\| \text{Imag}(\mathcal{F}(\mathbf{I}_{out})) - \text{Imag}(\mathcal{F}(\mathbf{I}_{gt})) \right\|_1, \quad (19)$$

where $\text{Real}(\cdot)$ and $\text{Imag}(\cdot)$ denote the real and imaginary parts in the Fourier domain, respectively.

4) *Perceptual Loss*: Perceptual loss supervises the output results through the pre-trained VGG19 network $\Phi(\cdot)$ to improve the perceptual quality:

$$L_{per} = \left\| \Phi(\mathbf{I}_{out}) - \Phi(\mathbf{I}_{gt}) \right\|_1. \quad (20)$$

The total loss function for training FDformer is defined as:

$$L_{stage2} = L_{pix} + \lambda_2 L_{fft} + \lambda_3 L_{per}, \quad (21)$$

where λ_2 is set to 0.1 and λ_3 is set to 0.01, empirically.

5) *Luminance Prediction Loss*: As shown in Fig. 4, MAR requires expect global luminance $\mathcal{R}$ to incorporate luminance information. In addition to allowing users to customize the global luminance, we also provide a straightforward method to obtain a reference global luminance. We leverage SEBlock in conjunction with a fully connected layer to infer $\mathcal{R}$, termed the Luminance Prediction Network (LPNet). Specifically, we formulate the luminance prediction as:

$$\mathcal{R}_{predict} = \varphi(\text{FC}(\text{Avg}_{1 \times 1}(\text{SE}(\mathbf{I}))))). \quad (22)$$

where $\mathbf{I}$ is the input low-light blurred image, $\mathcal{R}_{predict}$ signifies the anticipated global luminance, $\varphi(\cdot)$ denotes the sigmoid function, $\text{FC}(\cdot)$ denotes the fully connected layer, $\text{Avg}_{1 \times 1}(\cdot)$ denotes the global average pooling, and $\text{SE}(\cdot)$ represents a network of SEBlocks [63] with three different scales. We employ the MSE loss to train LPNet:

$$L_b = \left\| \mathcal{R}_{predict} - \mathcal{R}_{gt} \right\|_2. \quad (23)$$

IV. EXPERIMENTS

In this section, we present the training details of our method and evaluate its performance on publicly available datasets. Additionally, we conduct ablation experiments to demonstrate the validity of our design.

A. Datasets

Our FDN is trained on the LOL-Blur dataset [1], which consists of a total of 10,200 pairs of training data and 1,800 pairs of test data of low-light blurred and normal-light sharp images. To better demonstrate the performance of our method in real-world scenarios, we also tested the Real-LOL-Blur dataset [1], which includes 1,354 low-light blurred images captured in the real world without GT. Furthermore, we also evaluated FDN on the deblurring dataset GoPro [30] and the LLIE dataset LOL-v1 [64] to assess its performance under exclusively blur or low light degradation scenarios.

B. Implementation Details

We implemented our method using PyTorch on NVIDIA RTX A6000 GPUs and Intel Xeon E31200-series CPUs. Our FDN is trained with the Adam optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.9$). We apply a cosine annealing learning rate schedule, with a total of 100k iterations for MAR and 500k iterations for FDformer. The initial learning rate is set to 1×10^{-3} , and the minimum learning rate is 1×10^{-7} . Training data is obtained by applying random cropping, flipping, and rotation to original images, resulting in patches of size 128×128 pixels, with a batch size of 16. Our FDformer features a 3-scale encoder-decoder architecture with 6, 6, and 10 blocks at each scale. In FDSA, we use 8×8 pixels as the patch size and also set $\mathbf{W}_{amp}^n$ and $\mathbf{W}_{phase}^n$ in FDFFN to 8×8 .

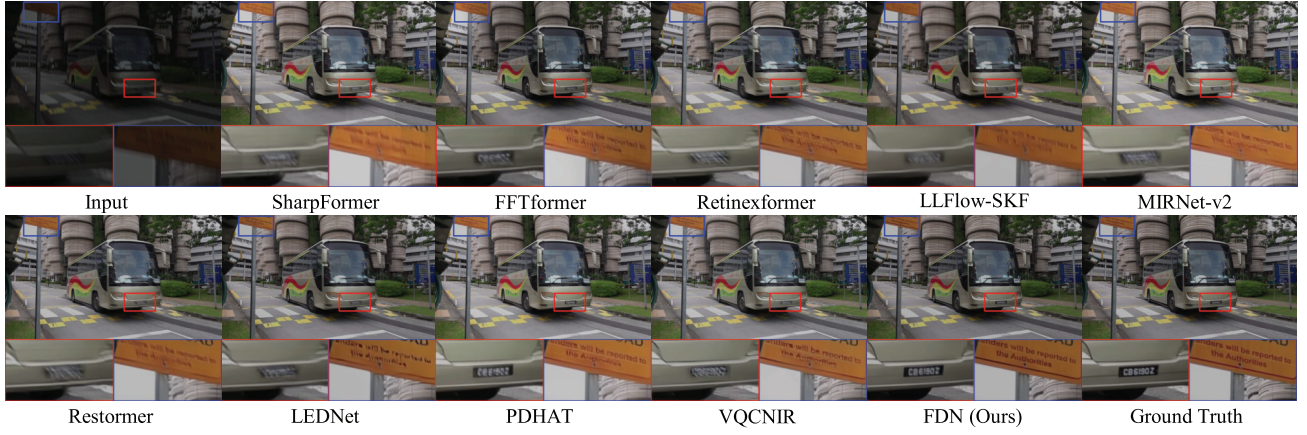


Fig. 9. Visual comparison results on the LOL-Blur dataset with methods retrained on the LOL-Blur dataset. The deblurring method performs better in restoring sharp edges, while the LLIE method excels in color restoration. Our method exhibits sharper edges.

TABLE I

QUANTITATIVE EVALUATION BETWEEN SINGLE NETWORK METHODS RETRAINED ON THE LOL-BLUR DATASET AND OUR METHOD. THE REPRESENTATION HIGHLIGHTED IN BOLD DEMONSTRATES THE BEST RESULTS ACROSS METRICS

Type	Methods	Params(M)	PSNR(dB)↑	SSIM↑	LPIPS↓	FID↓
Deblur	Sharpformer [35]	442.17	23.33	0.849	0.191	22.82
	FFTformer [37]	16.56	24.88	0.877	0.157	18.04
	FourLLIE [15]	0.12	20.16	0.726	0.330	42.66
LLIE	Retinexformer [29]	1.61	25.98	0.879	0.152	14.22
	LLFlow-SKF [28]	5.26	26.12	0.844	0.185	26.53
	MIRNet-v2 [65]	5.86	25.63	0.879	0.138	13.73
General	Restormer [62]	25.31	26.20	0.866	0.171	17.54
	LEDNet [1]	7.41	26.06	0.851	0.159	15.87
Joint	PDHAT [2]	7.83	26.71	0.885	0.175	12.90
	VQCNIR [3]	47.85	27.79	0.879	0.096	9.66
	FDN (Ours)	8.23	28.19	0.918	0.082	8.51

C. Results on LOL-Blur Dataset

We quantitatively compared our method with some widely used SOTA methods on the LOL-Blur test dataset. To demonstrate the necessity of designing models specifically for the joint tasks of LLIE and deblurring, we also selected several models specialized in either LLIE (FourLLIE [15], Retinexformer [29], LLFlow-SKF [28]), deblurring (FFTformer [37], Sharpformer [35]), or general image restoration (MIRNet-v2 [65], Restormer [62]). These models are retrained on the LOL-Blur dataset for a fair comparison. We use PSNR, SSIM, LPIPS [66], and FID [67] as the evaluation metrics for quantitative assessment.

We also consider the cascading of LLIE and deblurring models following [3]. We explore two cascading strategies: LLIE followed by image deblurring, and image deblurring followed by LLIE. Considering the lack of low-light sharp images and normal-light blurred images in the LOL-Blur dataset, retraining for cascading strategies is not feasible. For LLIE, we employ Retinexformer [29] and LLformer [68] pre-trained on the FiveK dataset [69]. For deblurring, we utilize FFTformer [37] and Restormer [62] pre-trained on the GoPro dataset [30].

1) *Single Network Methods*: Table I shows the quantitative comparison results on the LOL-Blur test dataset. Compared to the SOTA VQCNIR [3], our method reduces the parameter

TABLE II

QUANTITATIVE EVALUATION BETWEEN SIMPLE CASCADE METHODS AND OUR METHOD ON THE LOL-BLUR DATASET. THE REPRESENTATION HIGHLIGHTED IN BOLD DEMONSTRATES THE BEST RESULTS ACROSS METRICS

Type	Methods	PSNR(dB)↑	SSIM↑	LPIPS↓	FID↓
Deblur-LLIE	FFTformer→Retinexformer	15.05	0.627	0.402	62.83
	FFTformer→LLformer	17.89	0.627	0.341	49.82
	Restormer→Retinexformer	15.53	0.479	0.444	60.36
	Restormer→LLformer	18.53	0.633	0.434	48.16
	Retinexformer→FFTformer	14.85	0.418	0.492	58.31
LLIE-Deblur	Retinexformer→Restormer	15.17	0.450	0.447	61.91
	LLformer→FFTformer	17.38	0.545	0.489	95.63
	LLformer→Restormer	18.08	0.601	0.434	50.60
Joint	FDN (Ours)	28.19	0.918	0.082	8.51

count by 83.3% and outperforms it in terms of PSNR/ SSIM/ LPIPS/ FID by 0.4 dB/ 0.039/ 0.014/ 1.15. For fairness, we utilize the global luminance $\mathcal{R}_{predict}$ predicted by IPNet in the comparison. If we replace $\mathcal{R}_{predict}$ with $\mathcal{R}_{gt}$, our method achieves further improvements, reaching PSNR/ SSIM/ LPIPS/ FID values of 32.23 dB/ 0.931/ 0.077/ 7.83.

We also present the visual analysis of our method and other methods. As shown in Fig. 9, most methods perform poorly in handling severely blurred cases. In contrast, our method significantly outperforms others in restoring fine details and textures. Compared to non-decoupling methods VQCNIR, FDN excels in extracting blur-related degradation information, which provides a notable advantage in structural restoration. Moreover, VQCNIR has limitations in handling blurred edges, resulting in noticeable ghosting effects. Other LLIE methods lack mechanisms specifically designed for deblurring, making it difficult to restore sharp edges effectively. Although deblurring methods and general models can partially restore edges, they may introduce artifacts into the restored image.

2) *Simple Cascaded Methods*: Table II presents quantitative comparison results of our method with the simple cascaded

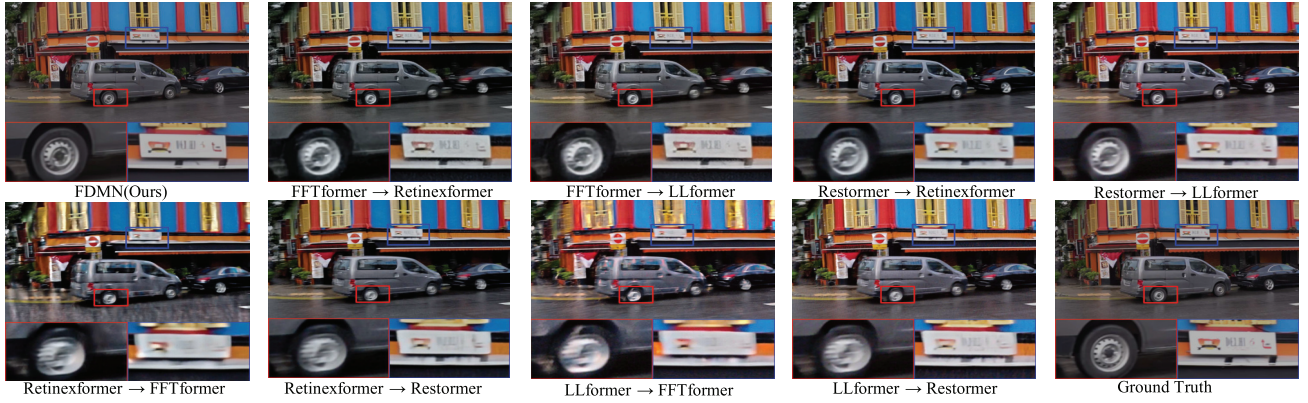


Fig. 10. Visual comparison results on the LOL-Blur dataset with simple cascade methods. In addition to our method, all other methods utilized officially pre-trained models. In the first row, the first image is our method, and the rest are results obtained by deblurring and then LLIE. In the second row, the last image represents GT, while the rest depict results obtained by LLIE and then deblurring. Our method demonstrates superior color restoration and sharp edges.

TABLE III

QUANTITATIVE EVALUATION BETWEEN SINGLE NETWORK METHODS RETRAINED ON THE REAL-LOL-BLUR DATASET AND OUR METHOD. THE REPRESENTATION HIGHLIGHTED IN BOLD DEMONSTRATES THE BEST RESULTS ACROSS METRICS

Type	Methods	NIQE↓	BRISQUE↓	PI↓
Deblur	Sharpformer [35]	4.995	38.62	4.847
	FFTformer [37]	4.710	37.55	4.485
LLIE	FourLLIE [15]	6.466	27.23	5.559
	Retinexformer [29]	5.874	45.07	5.488
	LLFlow-SKF [28]	5.275	40.89	5.228
General	MIRNet-v2 [65]	4.896	34.96	5.689
	Restormer [62]	5.737	40.91	5.306
Joint	LEDNet [1]	5.337	46.02	4.881
	PDHAT [2]	4.203	26.00	3.724
	VQCNIR [3]	4.599	35.92	3.805
	FDMN (Ours)	4.133	15.95	3.670

methods. From the quantitative metrics, it is evident that employing pre-trained models for simple cascading to address low-light blurred images results in significantly inferior quantitative performance compared to our approach.

Fig. 10 illustrates the visual results for comparison. Simple cascading methods result in color shifts. Deblurring before LLIE more effectively restores edges, whereas performing LLIE first yields superior noise reduction. However, in areas suffering severe blur or low light degradation, deblurring still remains challenging.

D. Results on Real-LOL-Blur Dataset

To better demonstrate the performance of our method in real-world scenarios, we test our model on the Real-LOL-Blur dataset, which is trained on LOL-Blur. We employed several widely used no-reference image quality assessment metrics, including NIQE [70], BRISQUE [71], and PI [72]. Additionally, we also compared the visualization performance of simple cascading methods on real datasets.

1) *Single Network Methods*: Table III presents the results of our method and other methods on the Real-LOL-Blur dataset, where our method outperforms others across all metrics.

Fig. 12 illustrates the visual effects of different methods. It can be observed that our method excels in restoring sharp

TABLE IV

QUANTITATIVE EVALUATION BETWEEN SIMPLE CASCADE METHODS AND OUR METHOD ON THE REAL-LOL-BLUR DATASET. THE REPRESENTATION HIGHLIGHTED IN BOLD DEMONSTRATES THE BEST RESULTS ACROSS METRICS

Type	Methods	NIQE↓	BRISQUE↓	PI↓
Deblur-LLIE	FFTformer → Retinexformer	5.260	43.52	4.932
	FFTformer → LLformer	4.206	19.41	4.186
	Restormer → Retinexformer	5.212	43.07	4.937
	Restormer → LLformer	4.346	19.04	4.253
	Retinexformer → FFTformer	5.120	46.22	4.909
LLIE-Deblur	Retinexformer → Restormer	5.113	43.34	4.909
	LLformer → FFTformer	4.265	18.95	4.155
	LLformer → Restormer	4.290	18.64	4.178
	FDMN (Ours)	4.133	15.95	3.670

image edges in real-world scenarios, while other methods struggle to restore information in severely blurred areas.

2) *Simple Cascaded Methods*: Table II presents a quantitative comparison between our proposed method and several simple cascading strategies on the Real-LOL-Blur dataset. As shown in the results, our method outperforms the cascading baselines across all evaluation metrics, demonstrating its effectiveness in handling complex degradations. Remarkably, certain cascade methods even outperform models trained on the LOL-Blur dataset. In addition, the visual comparisons shown in Fig. 11 further demonstrate the limitations of simple cascading methods. These methods often produce noticeable artifacts and color distortions, and they tend to struggle with denoising in regions containing severe degradation. In contrast, our method yields results with reduced noise, more accurate color restoration, and better preservation of structural details.

E. Results on LOL-v1 Dataset

To assess the performance of FDMN on solely low light degradation, we conducted experiments on the widely used LOL-v1 dataset [64]. We compared our method against nine

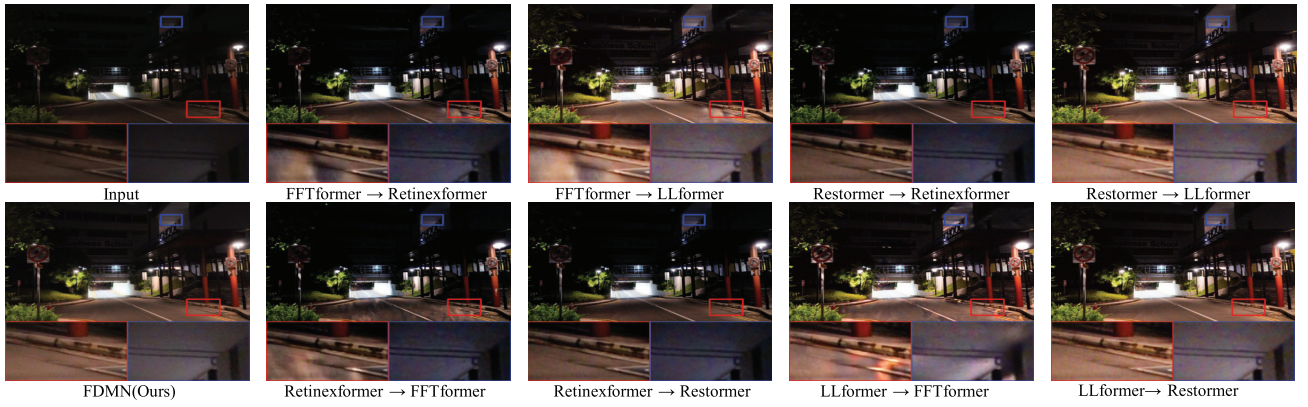


Fig. 11. Visual comparison results on the Real LOL-Blur dataset with simple cascade methods. In the first row, the first image is input, and the rest are results obtained by deblurring and then LLIE. In the second row, the first image represents our method, while the rest depict results obtained by LLIE and then deblurring. Our method demonstrates superior color and contrast restoration while avoiding overexposure in brighter regions.



Fig. 12. Visual comparison results on the Real LOL-Blur dataset. Our method exhibits sharper edges, particularly evident in the text regions.

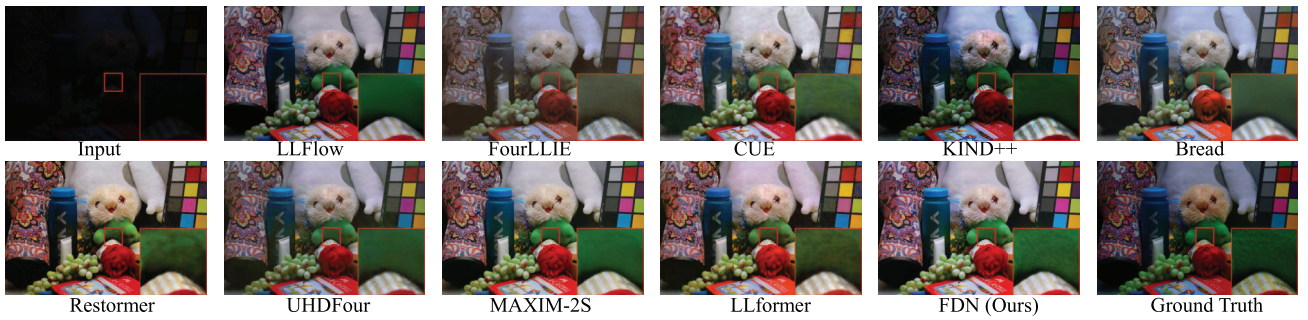


Fig. 13. Visual comparison results on the LOL-v1 dataset. Our method effectively preserves structural details and restores natural color.

SOTA LLIE methods: LLFlow [73], FourLLIE [15], CUE [74], KIND++ [75], Bread [76], Restormer [62], UHDfour [16], MAXIM-2S [77], LLformer [68]. All the aforementioned methods are trained on the same training set and evaluated on the testing set of the LOL-v1 dataset. Quantitative evaluation metrics include PSNR, SSIM, LPIPS, and FID as summarized in Table V. FDN achieves the best performance on PSNR (dB), LPIPS, and FID, with SSIM on par with MAXIM-2S. Visual comparisons are provided in Fig. 13, where prior methods exhibit varying degrees of over-smoothing or color shifts. In contrast, our method restores natural color and recovers finer details.

F. Results on GoPro Dataset

We also conducted experiments on the widely used GoPro dataset [30] to validate FDN on solely blur degradation. For this task, we converted FDformer into a four-scale U-Net [62] and set the number of blocks to 4, 6, 6, and 8 to enhance the capacity of the model. Since MAR is specific to LLIE, we remove it and feed FCAFFN with the amplitude and phase of the input images. The modified model is denoted as FDN-B. We compare our method against several deblurring methods: Deblur-GANv2 [78], DMPHN [79], MIMO-UNet+ [80], MPRNet [32], AdaDeblur [81], MSFNet [11], HINet [34],

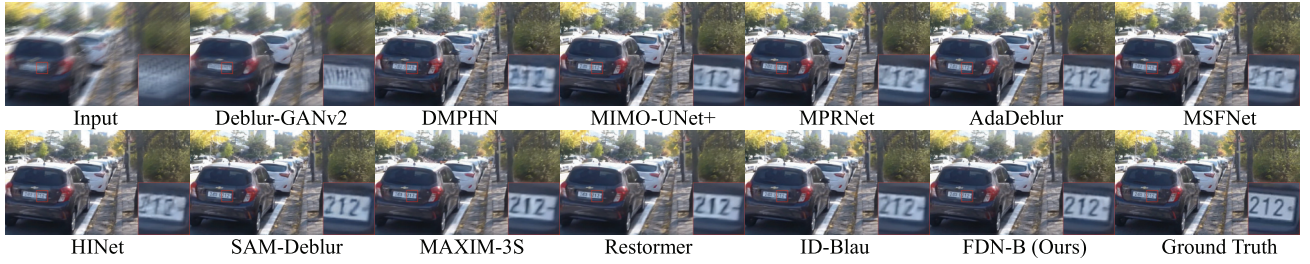


Fig. 14. Visual comparison results on the GoPro dataset. Our method exhibits sharper edges.

TABLE V

QUANTITATIVE EVALUATION ON LOL-V1 DATASET. THE REPRESENTATION HIGHLIGHTED IN BOLD DEMONSTRATES THE BEST RESULTS ACROSS METRICS

Method	PSNR(dB)↑	SSIM↑	LPIPS↓	FID↓
LLFlow [73]	19.34	0.892	0.144	67.74
FourLLIE [15]	20.82	0.867	0.173	70.50
CUE [74]	21.67	0.826	0.224	90.42
KIND++ [75]	21.80	0.881	0.158	106.26
Bread [76]	22.96	0.896	0.160	78.53
Restormer [62]	23.08	0.825	0.146	72.11
UHDFour [16]	23.10	0.872	0.147	56.93
MAXIM-2S [77]	23.43	0.910	0.099	40.82
LLformer [68]	23.65	0.873	0.169	75.50
FDN (Ours)	23.99	0.910	0.083	32.79

TABLE VI

QUANTITATIVE EVALUATION ON GoPRO DATASET. THE REPRESENTATION HIGHLIGHTED IN BOLD DEMONSTRATES THE BEST RESULTS ACROSS METRICS

Method	PSNR(dB)↑	SSIM↑	LPIPS↓	FID↓
Deblur-GANv2 [78]	29.08	0.879	0.117	15.65
DMPHN [79]	30.45	0.907	0.128	10.66
MIMO-UNet+ [80]	32.44	0.935	0.091	9.11
MPRNet [32]	32.66	0.938	0.089	8.90
AdaDeblur [81]	32.68	0.939	0.088	9.00
MSFNet [11]	32.72	0.938	0.088	8.35
HINet [34]	32.77	0.939	0.088	8.30
SAM-Deblur [82]	32.83	0.940	0.094	8.27
MAXIM-3S [77]	32.86	0.942	0.087	8.27
Restormer [62]	32.92	0.942	0.084	9.76
ID-Blau (MIMO-UNet+) [83]	32.93	0.941	0.088	7.91
FDN-B (Ours)	33.06	0.944	0.070	7.22

SAM-Deblur [82], MAXIM-3S [77], Restormer [62], ID-Blau (MIMO-UNet+) [83]. All the aforementioned methods are trained on the same training set and evaluated on the testing set of the GoPro dataset. Quantitative evaluation metrics include PSNR, SSIM, LPIPS, and FID as summarized in Table VI. Our method outperforms other comparative methods across all the evaluated metrics. Fig. 14 presents visual comparisons, demonstrating that our method restores sharper edges with fewer artifacts compared to other methods.

G. Results of Different $\mathcal{R}$

To demonstrate the flexibility of our method in customizing the luminance of results according to the pReferences of the users while maintaining restoration quality, we present results under several predefined $\mathcal{R}$ values shown in (Fig. 15(a)). As $\mathcal{R}$ varies, the primary changes occur in the amplitude component in response to different $\mathcal{R}$ values, while the phase

TABLE VII

QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS ON VARIOUS PARTS OF THE PROPOSED METHOD ON THE LOL-BLUR DATASET. THE REPRESENTATION HIGHLIGHTED IN BOLD DEMONSTRATES THE BEST RESULTS ACROSS METRICS

Component	Method	Params(M)	PSNR(dB)↑	SSIM↑	LPIPS↓
FDSA	MDTA+FDFFN	8.22	27.94	0.913	0.087
	FSAS+FDFFN	8.21	27.83	0.910	0.094
FDFFN	FDSA+GDFN	8.20	27.88	0.913	0.087
	FDSA+DFFN	8.22	27.97	0.914	0.086
	FDSA+MLP	8.23	27.94	0.912	0.095
	w/o phase filter	7.80	28.04	0.913	0.092
	w/o amplitude filter	7.80	27.90	0.911	0.087
FCAFFN	w/o FCA	8.19	27.64	0.905	0.092
	w/o modulation	8.22	27.93	0.917	0.091
FDN (Ours)		8.23	28.19	0.918	0.082

component remains largely stable, which confirms that our FDN effectively separates and adjusts the luminance through the amplitude component without degrading structural details through the phase component.

We also validate the accuracy of our luminance adjustment strategy in Fig. 15(b). The figure presents a scatter plot where each point in the plot represents one specific test image. Different colors indicate different $\mathcal{R}$ values. A reference line $\mathcal{L}_{real} = \mathcal{L}_{exp}$ is included for comparison, which demonstrates the reliability and consistency of our luminance customization approach across multiple $\mathcal{R}$ values. Additionally, the parameter count of LPNet is 0.14M, accounting for 5% of the total parameters.

H. Ablation Study

We conducted a series of ablation experiments to demonstrate the effectiveness of the components in our method.

1) *Improvements in FDSA*: Our proposed FDSA computes self-attention in the frequency domain. To validate its effectiveness, we compare it with the Multi-Dconv Head Transposed Attention (MDTA) [62] in the spatial domain and the Frequency domain-based Self-Attention Solver (FSAS) [37]. To ensure a fair comparison, we keep the number of parameters consistent by controlling the number of channels for $\mathcal{Q}$, $\mathcal{K}$, and $\mathcal{V}$. As shown in Table VII, both MDTA and FSAS yield lower performance than FDSA, with reductions in PSNR of 0.25 dB and 0.36 dB, respectively. This performance gap is due to the inability of MDTA and FSAS to decouple the degradations of low light and blur compared to FDSA, thus failing to fully leverage the correlations among different degradation factors.

Fig. 16 presents the attention maps generated by different self-attention mechanisms in Eq. 12. The heatmap of $\mathbf{Att}_{f1}$

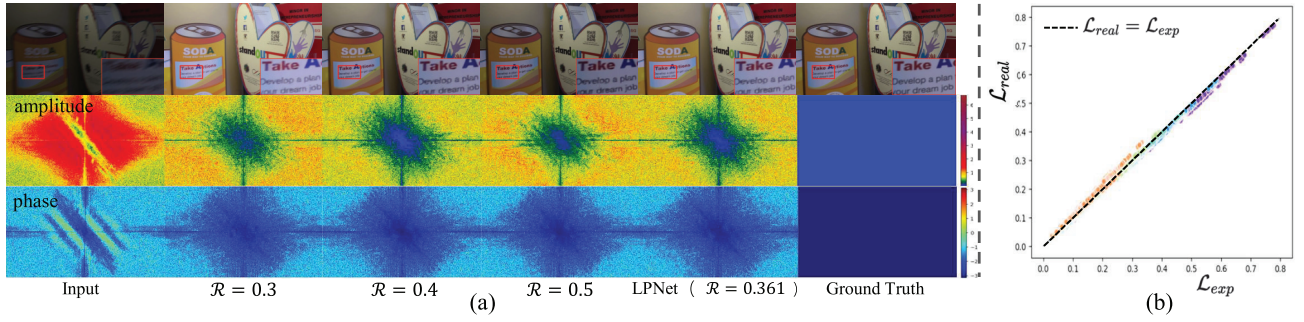


Fig. 15. (a) Visual comparison results on different $\mathcal{R}$ and their corresponding amplitude and phase. From left to right, the images are the input image, the output results with $\mathcal{R}$ of 0.3, 0.4, 0.5, and 0.361, and the GT on the far right. The result with $\mathcal{R} = 0.361$ is predicted by LPNet. Additionally, we visualized the differences in amplitude and phase between these images and the GT. (b) is the scatter plot of expected luminance $\mathcal{L}_{exp}$ vs. real luminance $\mathcal{L}_{real}$ on the LOL-Blur test dataset. Different colors represent different input $\mathcal{R}$ values.

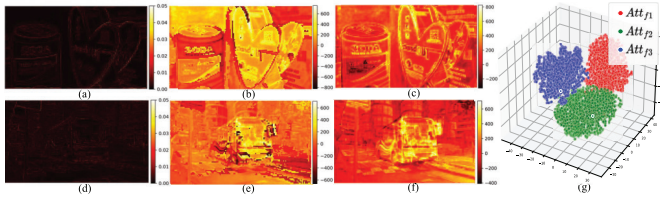


Fig. 16. The visualization of attention maps in FDSA. (a) and (d) represent the attention map Att_{f1} , (b) and (e) illustrate the attention map Att_{f2} , (c) and (f) demonstrate the attention map Att_{f3} . (g) shows the t-SNE results of these three attention maps from the smallest scale of FDN. The data is obtained from the LOL-Blur test dataset. The red, green, and blue circles represent Att_{f1} , Att_{f2} , and Att_{f3} , respectively. They exhibit distinct distributions.

reveals that the focal points are primarily concentrated on the edges associated with blurriness. While the heatmap of Att_{f2} shows that the higher attention values are concentrated in regions affected by luminance, in contrast to the edge-focused attention observed in Att_{f1} . Finally, the heatmap of Att_{f3} , which displays a more balanced attention distribution, targeting both edges and local regions. The t-SNE visualization in Fig. 16(g) demonstrates that the three attention maps are distinctly distributed, providing statistical evidence for our motivation to design different attention mechanisms for extracting different degradation representations.

2) *Improvements in FDFFN*: Similarly, to validate the effectiveness of FDFFN, we compare our approach with the spatial domain-based Gated-Dconv FN (GDFN) [62], and Frequency Domain-based FFN (DFFN). Additionally, we compared against the most commonly used MLP. Relative to these three different FFNs, our method achieved improvements in PSNR by 0.31 dB, 0.22 dB, and 0.25 dB, respectively. To further validate the necessity of the amplitude and phase filters in FDFFN, we conducted two sets of comparative experiments: one without the amplitude filter and another without the phase filter. While removing either the phase or amplitude filter can lead to a decrease in parameter count, the corresponding decrease in performance is also quite noticeable.

Additionally, we show the impact of the frequency branch and spatial branch (Fig. 17) in FDFFN by separately showing their amplitude and phase adjustments in the Fourier domain. In the amplitude component, both branches concentrate modifications in the mid-low frequency regions, which is consistent with our observation that mid-low frequencies encode most

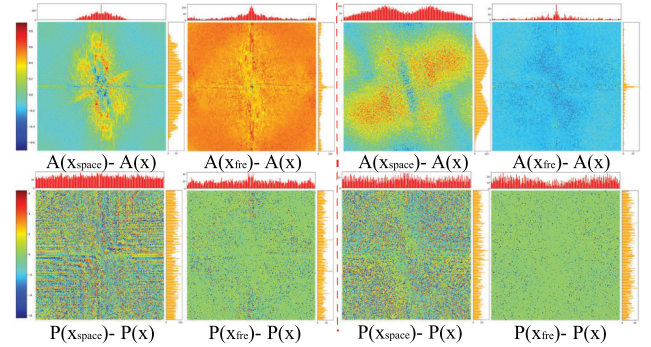


Fig. 17. Visualization of the frequency and spatial branches in FDFFN. The left and right sides of the red dotted line represent visualizations from different layers. To illustrate their effects, we calculate the transformation of the input feature x into x_{fre} and x_{space} in amplitude/phase. We also illustrate the variation across different frequency bands with histograms.

of the luminance feature related to statistical information. In the phase component, both branches introduce changes across all frequency bands, reflecting the ability to capture global structures (low frequencies) and fine details or edges (high frequencies) that are affected by blur degradation. Notably, the frequency branch demonstrates more sparse changes compared to the spatial branch, allowing for more targeted adjustments to specific frequency bands. This indicates that the two branches complement each other to learn distinct frequency characteristics. These visualizations demonstrate that FDFFN effectively disentangles and leverages amplitude and phase information to address different degradations.

3) *Improvements in the Component of FCAFFN*: In FCAFFN, we employed FCA and spatial modulation to integrate coarse-restored features. We individually removed these components to validate their effectiveness. Removing the FCA significantly degrades performance, as relying solely on spatial modulation fails to introduce coarse amplitude into the network effectively. Additionally, as discussed in Sec. III-D.3, removing the spatial modulation mechanism causes a mismatch between the restored coarse amplitude and the initial phase, resulting in a performance decline.

4) *Influence of MAR*: To demonstrate the impact of MAR on the result, we adjusted the zero-frequency component $|F(0,0)|$ of the degraded image to make it identical to the $|F(0,0)|$ of the amplitude restored by MAR. By modifying



Fig. 18. Comparison between the result without MAR and the result of FDN. Zoom in for a better view.

TABLE VIII

QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS OF THE TRAINING STRATEGY. THE REPRESENTATION HIGHLIGHTED IN BOLD DEMONSTRATES THE BEST RESULTS ACROSS METRICS

Method	PSNR(dB)↑	SSIM↑	LPIPS↓
Single-stage training	28.03	0.912	0.087
Coarse Alignment Loss → MSE Loss	28.09	0.912	0.089
w/o Amplitude Loss	28.11	0.915	0.086
w/o Frequency Loss	27.99	0.910	0.091
Luminance Prediction Loss as L1 Loss	28.05	0.916	0.085
FDN (Ours)	28.19	0.918	0.082

the $|F(0,0)|$ of both amplitudes to be identical, we can obtain results with the same global luminance, thereby facilitating the observation of the influence of other amplitude components on the results. As shown in Fig. 18, the absence of the luminance priors provided by MAR leads to noticeable luminance shifts and artifacts in the result. The quantitative results of PSNR/SSIM/ LPIPS/ FID are 25.79 dB/ 0.904/ 0.118 19.70. The parameter counts of MAR is 0.14M, accounting for only 1.7% of the total parameters.

5) *Improvement in the Training Strategy*: We further evaluate two proposed training strategies: (1) the necessity of the two-stage training scheme and (2) the contribution of individual loss functions. Specifically, we respectively replace the Coarse Alignment Loss with MSE Loss, the Luminance Prediction Loss (MSE Loss) with L1 Loss, and ordinate the Amplitude Loss and Frequency Loss during training, respectively. The results are presented in Table VIII.

Switching to single-stage training leads to degraded performance. The Coarse Alignment Loss and Amplitude Loss are essential for providing high-quality amplitude priors from MAR, which are critical for FDformer. Similarly, removing the Frequency Loss eliminates explicit frequency-domain constraints, resulting in further performance decline. Additionally, using MSE Loss for luminance prediction yields more accurate global luminance estimations by focusing on the mean values. In contrast, L1 Loss is biased toward the median values, reducing sensitivity to extreme values.

6) *Comparison With Other Luminance Priors*: We also compared our amplitude prior against other luminance priors, including the dark channel prior [84], the bright channel prior [85], and the Y channel prior, which represents luminance in YCbCr [86]. We present the results of $\mathcal{F}^{-1}(\beta F(u, v))$ with other luminance priors in Fig. 19. Our method isolates prior from phase, while the other luminance priors are phase-dependent, which hinders the effective decoupling of luminance information from phase component.

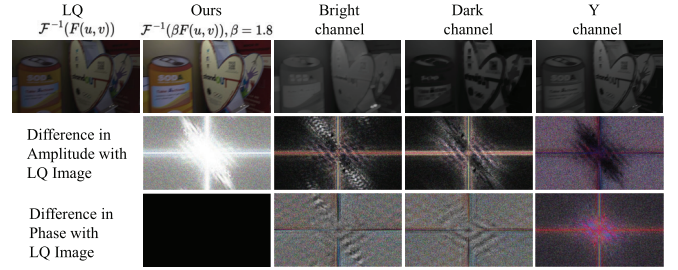


Fig. 19. Comparison between using $\mathcal{F}^{-1}(\beta F(u, v))$ as prior and other luminance priors.

V. CONCLUSION

We propose a method based on decoupling Fourier amplitude and phase for joint low-light image enhancement and deblurring. Our approach integrates two components: MAR, which focuses on restoring coarse amplitude, and FDformer, designed to extract and restore degradation. By effectively separating low light degradation and blur degradation, our method models these aspects independently in the amplitude and phase domains. This modeling strategy enables efficient restoration of image luminance, contrast, and sharp edges. Extensive experimental validation underscores the effectiveness of our approach. The potential of Fourier domain-based characterization of image degradation remains underexplored. Future research directions will explore more sophisticated methods for restoring complex degradations leveraging insights from the Fourier domain.

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